Small-Sample Classification for Hyperspectral Images With EPF-Based Smooth Ordering

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***Abstract*— Very limited training samples pose significant challenges for hyperspectral image (HSI) classification. To address this issue, small-sample learning methods based on classical machine learning or deep learning offer promising solutions. In this article, a novel two-stage learning-based small-sample classification framework is proposed for HSIs, termed edge-preserving features-based smooth ordering (EPFSO). In the proposed EPFSO, a self-training approach and two screening mechanisms are designed to iteratively learn newly labeled samples from a vast pool of unlabeled samples, thereby enhancing classification accuracies by incorporating these additional samples into the training set. The preprocessing step involves using edge-preserving filters to extract key features and generate low-dimensional feature images. Subsequently, all samples are ordered based on spectral similarity and spatial proximity, resulting in a smooth 1-D signal. In the case of limited labeled samples, a specialized self-training approach based on linear interpolation is utilized to iteratively learn newly labeled samples from unlabeled samples. This process continues until no further labeled samples are introduced, enabling gradual improvement in classification performance. In addition, two screening mechanisms are designed into the self-training process to strike a balance between the reliability and quantity of newly labeled samples. Finally, once a sufficient number of training samples are available, a majority voting mechanism is employed to efficiently classify the remaining samples. Experimental results on three open HSI datasets demonstrate that the proposed EPFSO framework outperforms several state-of-the-art methods, including six deep learning approaches. This validates the attractiveness of using EPFSO to address the challenges associated with limited labeled samples.**

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I. Introduction

HYPERSPECTRAL sensors can acquire very rich information with fine spectral resolution and spatial resolution, which makes it possible to implement an accurate classification of land covers. Whereas, the fine spectrum also produces problems for the land-cover classification of HSIs. The high dimensionality of spectral information and few labeled samples are two common challenges in HSI classification, because they can produce the Hughes phenomenon [1]. However, with the development of machine learning methods [2], [3], [4], especially deep learning ones, such as central attention network (CAN) [5], multiarea target attention (MATA) [6], SOT-NET [7], MDA-NET [8], and so on, the Hughes phenomenon was alleviated using some feature extraction techniques. Consequently, the classification performance of HSIs was enhanced to some extent. But, when few labeled samples are available, the improved classification accuracy is very limited [9]. It is well known that labeling samples is difficult and demanding, because it requires a lot of manpower and technology. Thus, small-sample classification of HSIs is an important and urgent task that has attracted widespread attention in HSI processing.

In numerous HSI classification tasks, novel machine learning and deep learning algorithms have made great progress in the case of relatively sufficient labeled samples. In some open HSI scenes, classification accuracies have achieved a near-perfect result, close to 100%. Considering the requirements of practical applications (further improvements in the spectral resolution and spatial resolution, and the cost of manual labeling), small-sample classification is becoming the focus of HSI processing. This is mainly due to the Hughes phenomenon, which is produced by high-dimensional spectral information and fewer labeled samples. Feature extraction can effectively overcome the high dimensionality of spectral information. Meanwhile, since HSIs have rich spatial structure and texture information, many spatial feature extraction methods are designed to improve classification accuracies. However, the additional introduction of spatial information increments the curse of dimensionality and exacerbates the Hughes phenomenon when very few labeled samples are available. Furthermore, too few samples inherently limit the

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generalization ability of some classifiers [9], [10], such as SVM, random forest (RF), multiple linear regression (MLR), convolutional neural network (CNN), Transformer, and so on, thereby leading to poor classification performance. Thus, learning more new labeled samples to improve classification accuracies is our main research task.

Small-sample learning is mainly based on classical machine learning and deep learning algorithms, incorporating some new learning mechanisms. Semisupervised learning and active learning are two categories of small-sample learning. One of the most significant differences between semisupervised learning algorithms and supervised ones is the utilization of unlabeled samples. When few labeled samples are available, semisupervised algorithms can learn new samples from unlabeled samples for data augmentation [11], [12], [13]. Active learning [14], [15] is a machine learning algorithm that can achieve greater accuracies with few labeled samples if it is allowed to choose the training samples from which the learning is based. An active learner may ask queries in the form of unlabeled samples to be labeled by a human annotator. The notable characteristic of active learning is the need for human participation. Besides, generative models have been also used for small-sample learning by generating new training samples based on labeled samples [9]. This article mainly focuses on a semisupervised learning method for small-sample classification of HSIs.

In recent years, various small-sample learning methods have been presented for land-cover classification of HSIs [16], especially some deep learning methods [17]. For example, a semisupervised CNN with a ladder network (S2DCNN) was presented for land-cover classification of HSIs [18]. Gao et al. [19] designed a small-sample learning method UM2L for HSI classification. A novel deep small-sample learning method DCFSL for HSI classification is presented by Li et al. [20]. Lu et al. [21] presented a new band-grouping-guided multiattention method (GMA-Net) for the performance promotion of spectral–spatial feature learning in HSI classification. Yang et al. [22] proposed a semisupervised dynamic multiscale graph convolutional network classifier (DMSGer) for accurate HSI classification. Recently, Lu et al. [23] proposed a novel dual-stream class-adaptive network (DSCA-Net) with semisupervised learning for HSI classification. These methods adopt various deep learning networks to extract effective features and get excellent performance with a few labeled samples. Nevertheless, these deep learning models often involve a considerable number of tunable parameters. By comparison, classic machine learning methods boast exceptional advantages in the small-sample classification of HSIs. Moreover, they have few parameters to be determined and can be trained and tested quickly. For example, Jia et al. [24] designed a novel decision fusion method combining local binary pattern (LBP) and superpixel segmentation for land-cover classification of HSIs. Inspired by this, Zheng et al. [25] designed a small-sample classification method using a distance-weighted linear regression classifier instead of decision fusion. Zhang et al. [26] presented a semisupervised learning method with linear iterative cluster segmentation for HSI classification. Small-sample learning methods based on classic machine learning theory have also achieved relatively good classification accuracies with limited samples. However, regardless of whether they are deep models or classical models, the classification performance remains far from the expected level. Therefore, further research and development of more efficient small-sample learning methods are needed.

Numerous studies have proved that the utilization of spatial features significantly enhances the accuracies of land-cover classification of HSIs [3], [27]. Therefore, research on effective spatial information extraction methods is crucial for HSI classification tasks. To date, several effective techniques have been presented for extracting spatial properties, such as shape, texture, edge, and so on. In the initial stages of these techniques, some statistical features, such as mean and variance, are used for mining spatial features of HSIs [28], [29]. Widely used tools for HSI land-cover classification, including morphological profiles (MPs), attribute filters (AFs), extended MP (EMP), extended multiattribute profiles (EMAPs), and various variants, were presented in [30], [31], and [32], respectively. LBP has also been introduced into hyperspectral image processing [24], [33]. This tool aims at extracting spatial features of images according to neighborhood structure relationships. Superpixel segmentation [25], [34] is a class of methods developed in recent years and widely used for mining spatial features of HSIs. With the flourishing of deep learning technology, some deep feature extraction methods, such as 2D-CNN and 3D-CNN [18], [20], [35], were introduced for spectral–spatial classification tasks. The edge-preserving filter is a very popular tool for spatial feature extraction in HSI classification [36], [37]. The filter effectively extracts texture and edge features from HSIs without blurring edges. It is well suited for HSI feature extraction because of the consistent land-cover distribution. So, we use the edge-preserving filter to get the initial features.

It is well known that high-dimensional data are usually sparsely distributed over some manifold [38], [39], [40]. Thus, developing low-dimensional embedding algorithms can effectively mine the distribution characteristics of data in the manifold. Some classical embedding algorithms [3], [41], such as random projection, locally linear embedding, linear discriminant analysis, nonparametric weighted feature extraction, and some extended methods, have been extensively used for extracting low-dimensional features in HSI processing. However, these methods make it difficult to produce discriminative features directly with little prior information. Thus, we will design a simple and effective low-dimensional embedding method based on smooth ordering and the shortest paths, which aims at mining the arrangement and distribution of samples over some manifold by exploiting spectral similarity and spatial proximity.

In this work, a new small-sample classification framework, EPFSO, is presented for HSI classification. This method can achieve the fast and high-precision classification of HSIs by effectively overcoming the problems of high data dimensionality and few samples. In short, EPFSO can be described

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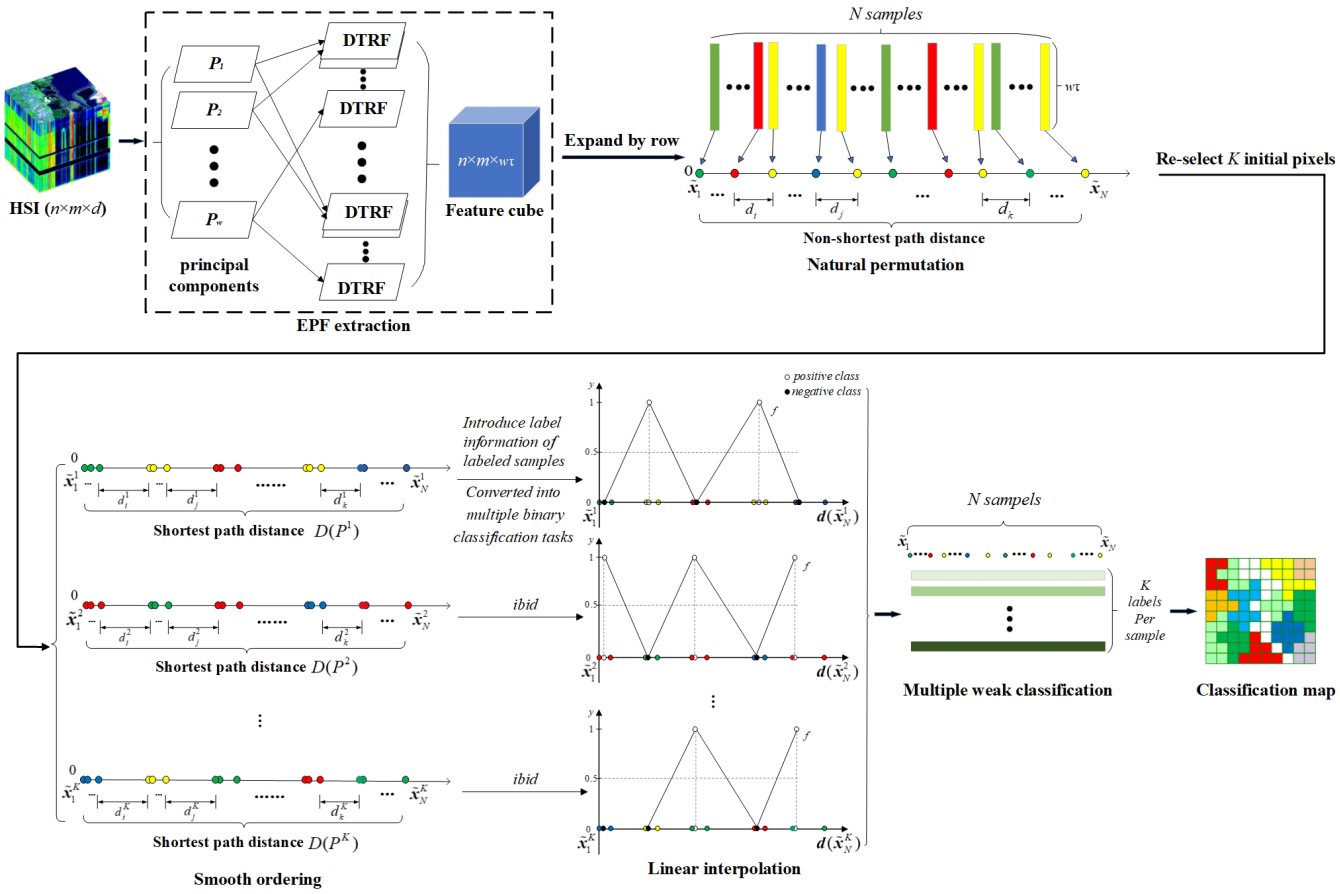


Fig. 1. Main flowchart of EPFSO for HSI classification, including five modules, i.e., EPF extraction used for spectral–spatial features, natural permutation used for a 1-D signal, smooth ordering based on EPF used for feature reordering, linear interpolation used for binary classification, and multiple weak classification results used for label learning.

below. First of all, we introduce the edge-preserving filters to extract the edge-preserving features (EPFs) as the initial features. Second, a smooth ordering operator is implemented on the EPF to obtain the shortest path of samples. Then, a simple linear interpolating method is used to classify samples by the distance curve of the shortest path. Next, we design a self-training strategy and two screening mechanisms that are used to iteratively learn newly labeled samples through a repeated process of smooth ordering and interpolation classification. The principle combines the results of multiple weak classifications produced by the interpolation to generate reliable new labeled samples. Finally, based on the augmented training set, the remaining samples are classified by the common majority vote mechanism according to the results of multiple weak classifications. The main flowchart for EPFSO is illustrated in Fig. 1. The main contributions of the work can be summed up as follows.

1) A novel two-stage learning framework, EPFSO, is proposed for high-accuracy classification for HSIs, which can alleviate the problems of high data dimensionality and limited training samples in a semisupervised mode. A specialized self-training strategy is designed in this proposed framework to iteratively learn newly labeled samples through a repeated process of smooth ordering and interpolation classification.
2) Two different screening mechanisms, i.e., the error-free mechanism and the fault-tolerant mechanism for a single error, are designed to overcome the poor reliability of new training samples and the insufficient number of new samples. The error-free mechanism can ensure that EPFSO can learn the newly labeled samples with almost 100% accuracy, but the number of labeled samples is limited. On the other hand, the fault-tolerant mechanism for a single error allows EPFSO to acquire a sufficient number of newly labeled samples, albeit with a slight compromise in accuracy. A mixture of both mechanisms can achieve a balance between the quantity and accuracy of newly labeled samples.
3) To classify the remaining samples and avoid repeated ordering, we label newly labeled samples based on the existing sequences and employ linear interpolation and majority voting mechanisms to achieve the final classification. Furthermore, we conduct both qualitative and quantitative assessments for the classification performance of the proposed EPFSO on three representative HSI datasets. The experimental outcomes demonstrate the remarkable superiority of the EPFSO compared with both traditional small-sample learning methods and other cutting-edge deep learning methods.

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The rest of this work is structured as follows. Section II mainly introduces edge-preserving filters and presents smooth ordering based on EPF. Section III analyzes the principle of the self-training approach based on linear interpolation and describes two different screening mechanisms. Section IV summarizes the small-sample classification based on an AdaBoost framework and analyzes the time complexity of EPFSO. Section V gives experimental results and some analysis between EPFSO and other comparative methods. Finally, this work is concluded with some remarks in Section VI.

II. Smooth Ordering Based on EPF

Edge-preserving filters can extract spectral–spatial features EPF of HSIs, and yet smooth ordering can implement the lower dimensional embedding transformation of EPF. This method aims to realize the sequential arrangement of samples on the path according to spectral similarity and spatial homogeneity and obtain the shortest path. This can lay the foundation for learning newly labeled samples in the semisupervised strategy.

Some common notations are first defined as follows. Given an HSI $\boldsymbol{X} \in \mathbb{R}^{n\times m\times d}$, including $N = n \times m$ samples $\{\boldsymbol{x}_i, y_i\}_{i=1}^{N}$. Here, $\boldsymbol{x}_i \in \mathbb{R}^d$ is a vector of the ith sample, whose label is $y_i \in \boldsymbol{\Omega} = \{1, 2, \ldots, C\}$. Next, edge-preserving filters [36] are briefly described as follows. The first w principal components $\boldsymbol{P}_1, \boldsymbol{P}_2, \ldots, \boldsymbol{P}_w \in \mathbb{R}^{n\times m}$ with the cumulative variance contribution rate over 99% are extracted from the cube image $\boldsymbol{X}$ by principal component analysis (PCA). Based on the first w principal components, a single edge-preserving feature $\boldsymbol{F}$ is built by domain transform filtering operation DTRF$(\cdot)$ [42] with a smoothing parameter pair (δ_s, δ_r) as follows:

$$\begin{aligned}\boldsymbol{F}(\delta_s, \delta_r) = \{&\mathrm{DTRF}(\boldsymbol{P}_1, \delta_s, \delta_r)\\ &\mathrm{DTRF}(\boldsymbol{P}_2, \delta_s, \delta_r), \ldots, \mathrm{DTRF}(\boldsymbol{P}_w, \delta_s, \delta_r)\}. \quad (1)\end{aligned}$$

Furthermore, by integrating multiple EPFs $\boldsymbol{F}$ with different smoothing parameter pairs in a stacked manner, the low-dimensional spectral–spatial features $\tilde{\boldsymbol{X}}$ of an HSI can be obtained as follows:

$$\tilde{\boldsymbol{X}} = \left\{\boldsymbol{F}\left(\delta_s^1, \delta_r^1\right), \boldsymbol{F}\left(\delta_s^2, \delta_r^2\right), \ldots, \boldsymbol{F}\left(\delta_s^\tau, \delta_r^\tau\right)\right\}. \quad (2)$$

Here, τ denotes the number of smoothing parameter pairs. According to the resulting spectral–spatial features $\tilde{\boldsymbol{X}}$, a lower dimensional embedding transformation method, EPFSO, based on smooth ordering is built to simply realize the aggregation and separation of samples on a 1-D permutation. This method is mainly established based on the feature similarity measure and spatial neighborhood relationship of samples.

Note that the EPFs $\tilde{\boldsymbol{X}}$ still form a feature cube with the size $n \times m \times w\tau$ ($w\tau \ll d$). Then, the cube $\tilde{\boldsymbol{X}}$ can be spread in row as $\{\tilde{\boldsymbol{x}}_{m*(i-1)+j}\}_{i=1,j=1}^{i=n,j=m}$ according to the order in the matrix, where $\tilde{\boldsymbol{x}}_{m*(i-1)+j} \in \mathbb{R}^{w\tau}$ denotes the feature of the pixel at the position (i, j), that is,

$$\tilde{\boldsymbol{X}} \xrightarrow{\text{spread-in-row}} \left\{\tilde{\boldsymbol{x}}_1, \tilde{\boldsymbol{x}}_2, \ldots, \tilde{\boldsymbol{x}}_N\right\}. \quad (3)$$

This method of directly converting a 2-D matrix into a 1-D vector does not guarantee the similarity between adjacent pixels and destroys spatial smoothness. To deal with this problem, a low-dimensional embedding method, EPFSO, is designed for the reorder of $\{\tilde{\boldsymbol{x}}_{m*(i-1)+j}\}$ based on the feature similarity measure and spatial neighborhood relationship of samples in the original matrix. Thus, the smooth ordering method includes two main steps, i.e., feature similarity measure and neighborhood selection. For simplicity and efficiency, Euclidean distance is adopted for the former, while the latter generally adopts a fixed neighborhood. This local smooth ordering strategy boasts two significant advantages in contrast with global smooth ordering. One is decreasing the computing cost, because the global search in the whole HSI at each step needs to take too much time. Another is the use of a spatial structure, because large homogeneous regions exist in HSIs. The goal of EPFSO is to get a highly smooth permutation by minimizing the total distance measure. Furthermore, by setting the initial point as the origin, this highly smooth permutation can be regarded as a 1-D signal spaced by distance measure. The specific steps of EPFSO are as follows.

In EPFSO, the choice of the initial point is very important for smooth ordering. Different initial points produce different permutations. In general, the initial point $\tilde{\boldsymbol{x}}_1^*$ is randomly selected from the set $\{\tilde{\boldsymbol{x}}_i\}_{i=1}^{N}$. Let P be a smooth ordering operator. The shortest path optimization problem based on EPFs $\{\tilde{\boldsymbol{x}}_i\}_{i=1}^{N}$ is established as follows:

$$\begin{aligned}&\min_P \; D(P) = \sum_{k=1}^{N-1} \left\|\tilde{\boldsymbol{x}}_{k+1}^* - \tilde{\boldsymbol{x}}_k^*\right\|_2\\ &\text{s.t.} \quad \left(\tilde{\boldsymbol{x}}_1, \tilde{\boldsymbol{x}}_2, \ldots, \tilde{\boldsymbol{x}}_N\right) \xrightarrow{P} \left(\tilde{\boldsymbol{x}}_1^*, \tilde{\boldsymbol{x}}_2^*, \ldots, \tilde{\boldsymbol{x}}_N^*\right) \quad (4)\end{aligned}$$

where $\|\cdot\|_2$ represents the Euclidean distance metric. The operator P aims to reorder this raw permutation $(\tilde{\boldsymbol{x}}_1, \tilde{\boldsymbol{x}}_2, \ldots, \tilde{\boldsymbol{x}}_N)$ and produce a new smooth permutation $(\tilde{\boldsymbol{x}}_1^*, \tilde{\boldsymbol{x}}_2^*, \ldots, \tilde{\boldsymbol{x}}_N^*)$. This is a classic shortest-path problem, i.e., how to minimize the total distance. Nevertheless, numerous samples in an HSI can lead to high computational costs, coupled with the existence of some large homogeneous regions. Local smooth ordering instead of global smooth ordering has become the best choice, which also means finding a suboptimal solution instead of the optimal one. In fact, this method aims to make samples from the same class order in adjacent positions and allow a little error tolerance. Thus, the solution idea of optimization problem (4) is given as follows.

According to the initial point $\tilde{\boldsymbol{x}}_1^*$, the neighborhood of $\tilde{\boldsymbol{x}}_1^*$ with radius r and without center is defined as $\dot{U}_r(\tilde{\boldsymbol{x}}_1^*)$ in the feature cube $\tilde{\boldsymbol{X}}$. Next, the second point $\tilde{\boldsymbol{x}}_2^*$ should be selected from the neighborhood $\dot{U}_r(\tilde{\boldsymbol{x}}_1^*)$ based on the shortest distance. This process can continue until all samples are completed without repetition. However, a sharp problem is that the front part of the permutation is smooth, but its end distance changes drastically. The main reason is that the points at the head of the permutation can be followed by the next points with a smaller distance, but the room for subsequent selection at the end of the permutation is getting smaller and smaller. The

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conditional random selection principle can effectively alleviate this problem. The main methodology is as follows:

$$\begin{aligned} \tilde{\boldsymbol{x}}_2^* &= \mathbf{1}_{\{q>p\}}\tilde{\boldsymbol{x}}_2^1 + \mathbf{1}_{\{q\le p\}}\tilde{\boldsymbol{x}}_2^2, \\ \text{where} \quad q &= \frac{1}{1+\exp\left(\frac{\|\tilde{\boldsymbol{x}}_2^1-\tilde{\boldsymbol{x}}_1^*\|_2-\|\tilde{\boldsymbol{x}}_2^2-\tilde{\boldsymbol{x}}_1^*\|_2}{\epsilon}\right)} \\ \tilde{\boldsymbol{x}}_2^1 &= \arg\min_{\boldsymbol{x}\in\dot{U}_r(\tilde{\boldsymbol{x}}_1^*)} \left\|\tilde{\boldsymbol{x}}-\tilde{\boldsymbol{x}}_1^*\right\|_2 \\ \text{and} \quad \tilde{\boldsymbol{x}}_2^2 &= \arg\min_{\boldsymbol{x}\in\dot{U}_r(\tilde{\boldsymbol{x}}_1^*)/\tilde{\boldsymbol{x}}_2^1} \left\|\tilde{\boldsymbol{x}}-\tilde{\boldsymbol{x}}_1^*\right\|_2. \end{aligned} \tag{5}$$

Here, $\mathbf{1}_{\{\cdot\}}$ is a Boolean function, p is randomly selected from [0.5, 1], and the bandwidth parameter ϵ is positive. Obviously, these two candidate points $\tilde{\boldsymbol{x}}_2^1$ and $\tilde{\boldsymbol{x}}_2^2$ are the most similar and second most similar points to the center point $\tilde{\boldsymbol{x}}_1^*$, respectively. In the case that the two distances are similar, the above method tends to select the second most similar point. Otherwise, the most similar point is selected as the following point in the permutation. After obtaining the new point $\tilde{\boldsymbol{x}}_2^*$, its following point $\tilde{\boldsymbol{x}}_3^*$ can be searched in its neighborhood $\dot{U}_r(\tilde{\boldsymbol{x}}_2^*)$ by the conditional random selection principle (5). Proceeding in turn, we can achieve an ordered permutation of all these points. In addition, it should be noted that when there is only one candidate point in the neighborhood, we directly select the point, and when there is no candidate point in the neighborhood, we extend to select the most similar point globally. After finishing the process, the new permutation $(\tilde{\boldsymbol{x}}_1^*, \tilde{\boldsymbol{x}}_2^*, \ldots, \tilde{\boldsymbol{x}}_N^*)$ is produced starting from this initial point $\tilde{\boldsymbol{x}}_1^*$. Next, the shortest path distances in this new permutation are defined as follows:

$$d\left(\tilde{\boldsymbol{x}}_i^*\right) = \sum_{j=1}^{i-1} \left\|\tilde{\boldsymbol{x}}_{j+1}^* - \tilde{\boldsymbol{x}}_j^*\right\|_2, \quad i = 2, \ldots, N. \tag{6}$$

In particular, it should be noted that $d(\tilde{\boldsymbol{x}}_1^*) = 0$. In this way, we achieve extreme dimensionality reduction of features, embedding all pixels along the shortest paths. At the same time, a series of nodes with the coordinate $\{(d(\tilde{\boldsymbol{x}}_i^*), y_i^*)\}_{i=1}^N$ are constructed in the coordinate system. Next, a self-training strategy based on interpolation classification is designed for learning newly labeled samples.

III. Self-Training Strategy and Two Screening Mechanisms

Different initial points produce different shortest paths and result in different node sets. For each set of nodes $\{(d(\tilde{\boldsymbol{x}}_i^*), y_i^*)\}_{i=1}^N$, labels of samples (unlabeled) can be quickly produced by linear interpolation, based on the label information of labeled samples (initial training set). The classification task with C classes is transformed into C binary classification tasks by using the one-vs-all strategy. In the binary classification task, when considering the cth class ($c = 1, 2, \ldots, C$) as the positive class and the others as the negative class, these nodes $\{(d(\tilde{\boldsymbol{x}}_{i_l}^*), \varphi_{i_l})\}_{l=1}^L$ corresponding to labeled samples $\{(\tilde{\boldsymbol{x}}_{i_l}^*, y_{i_l}^*)\}_{l=1}^L$ are reset as follows:

$$\varphi_{i_l} = \begin{cases} 1, & \text{if} \quad y_{i_l}^* = c \\ 0, & \text{if} \quad y_{i_l}^* \neq c \end{cases} \quad \text{where} \quad l = 1, \ldots, L. \tag{7}$$

The purpose of the above transformation (7) is to find out the positions of the cth labeled samples in the new permutation, then set their y-axis values as 1, and assign 0 to the other labeled samples. Based on the known nodes $\{(d(\tilde{\boldsymbol{x}}_{i_l}^*), \varphi_{i_l})\}_{l=1}^L$, we can construct an interpolation function f_c, so that the following condition

$$f_c\left(d\left(\tilde{\boldsymbol{x}}_{i_l}^*\right)\right) = \varphi_{i_l}, \quad l = 1, \ldots, L \tag{8}$$

is met. Once we have obtained the interpolation function f_c, we can find out all samples that may be classified into the cth class by substituting their distances $\{d(\tilde{\boldsymbol{x}}_i^*)\}_{i=1}^N$, i.e.,

$$y_i^* = c, \quad \text{if} \quad f_c\left(d\left(\tilde{\boldsymbol{x}}_i^*\right)\right) > 0.5. \tag{9}$$

All classes (i.e., $c = 1, 2, \ldots, C$) are considered in turn once by (7)–(9), and we can get the labels of all samples. Although some samples are given multiple labels (labels of these samples are temporarily marked as 0), only those samples assigned a single label $c (c = 1, 2, \ldots, C)$ are considered. These samples with a single label are regarded as achieving preliminary correct classification results.

The classification results obtained by implementing interpolation classification on a permutation may not be very reliable. Yet, we can choose K different initial points $\{\tilde{\boldsymbol{x}}_1^{*k}\}_{k=1}^K$ and then get K permutations by using the proposed EPFSO. For each sample, we can obtain a label from one permutation. Whereupon, K labels for each sample can be obtained from the K permutations, taking the ith sample in the natural permutation

$$\boldsymbol{y}_i = \left[y_{i_1}, y_{i_2}, \ldots, y_{i_K}\right]^T. \tag{10}$$

According to the results (10), a more reliable classification label for the ith sample can be produced in an AdaBoost framework based on multiple interpolation classification. Here, two mechanisms are built to produce final classification results. One is an error-free mechanism, i.e.,

$$y_i = c, \quad \text{if} \quad \exists c \in \boldsymbol{\Omega}, \quad \text{s.t.} \quad \boldsymbol{y}_i \equiv c. \tag{11}$$

Another is a fault-tolerant mechanism for a single error, i.e.,

$$y_i = c, \quad \text{if} \quad \exists c \in \boldsymbol{\Omega}, \quad \text{s.t.} \sum_{k=1}^{K} \mathbf{1}_{\{y_{i_k}=c\}} \geq K - 1. \tag{12}$$

Obviously, the former is more stringent than the latter. Based on the two mechanisms, a specialized self-training method is designed to iteratively learn newly labeled samples. The former is used in the first stage, while the latter is in the second stage.

For the K permutations, the error-free mechanism (11) is first used in (10) to obtain some new labeling samples with high reliability. By enlarging the training set with these new labeling samples, the interpolation classification processing (7)–(9) is again implemented on the above K permutations to obtain the new K labels, such as (10), for each sample. Then, the fault-tolerant mechanism for a single error (12) is used in new K labels to obtain more new labeling samples. Although the accuracy of these newly added labeled samples in the 2nd stage is inferior to that in the 1st stage, it also boasts excellent performance as K increases. Moreover, the number

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of newly added samples in the 2nd stage is much more than that in the 1st stage.

By introducing the newly labeled samples, a sufficient number of labeled samples are obtained for testing the remaining samples. To quickly classify these remaining samples, interpolation operations (7)–(9) are again implemented on the K permutations to obtain the K labels for each remaining sample. Finally, majority voting is used to finish the classification task.

IV. Small-Sample Classification Based on the Proposed EPFSO

To tackle the small-sample classification tasks of land covers in HSIs, this work presents a fast and efficient semisupervised learning framework with a self-training approach and two screening mechanisms. The newly labeled samples are learned in a self-training framework, which produces a reliable label by integrating the results of multiple weak classifiers for each sample. The idea behind this model is quite straightforward to implement. First, edge-preserving filtering is adopted for dimensionality reduction and spectral–spatial feature extraction of HSIs. Then, smooth ordering based on the shortest path distance is proposed for further feature embedding and order, which results in a permutation starting from some initial point. Next, the simple weak classifier based on linear interpolation is designed for the preliminary classification of unlabeled samples. An AdaBoost algorithm based on linear interpolation is built for enhancing the reliability of the labels of some test samples and learning newly labeled samples. The hybrid use of the two screening mechanisms achieves a balance between the number and accuracy of newly labeled samples. Finally, a new semisupervised classification framework, EPFSO, is designed to implement the small-sample classification of land covers in HSIs. The proposed EPFSO utilizes the self-training strategy and two screening mechanisms to continuously expand the training set. When sufficient training samples are available, the remaining samples can be classified with high accuracies by using a simple classifier.

Next, the computational complexity of EPFSO is briefly evaluated. The computational cost mainly results from feature extraction by edge-preserving filtering, feature embedding based on smooth ordering, and linear interpolation. Here, the edge-preserving filtering enables a high-quality real-time $O(N)$ algorithm [43]. The computational cost of smooth ordering is determined by the sample size N (in this whole map), the dimensionality of EPF $w\tau$, and the radius of neighborhood r. Thus, the number of operations required is about $O(w\tau N(2r+1)^2)$ for a permutation. The linear interpolation also enables fast real-time processing with $O(NC)$, related to the sample size N and the number of classes C. Thus, the number of operations for K permutations is about $O(Kw\tau N(2r+1)^2 + KNC)$. Combining the feature extraction by edge-preserving filtering, the total computational complexity of EPFSO requires $O(N+Kw\tau N(2r+1)^2+KNC)$, approximate $O(Kw\tau Nr^2)$ due to the number of classes is usually far less than $w\tau r^2$. Actually, only the target classes are considered, excluding the background class in an HSI, so the number of samples is less than N. Besides, due to the nonrepetitive order, the number of samples in the neighborhood would be less and less, and the number of operations is actually under $O(Kw\tau Nr^2)$.

V. Experimental Results and Analysis

In this section, we compare EPFSO with some advanced methods on three benchmark HSI datasets in the small-sample classification tasks, as follows: 1) Indian Pines (IPs); 2) Pavia University (PU); and 3) WHU-Hi-HongHu (HongHu). The methods used for comparison are SVM, PCA-EPFs [42], ULBP-SPG [24], STSE-DWLR [25], S2DCNN [18], UM2L [19], DCFSL [20], DSCA-Net [23], GMA-Net [21], and DMSGer [22], where SVM and PCA-EPFs are two benchmark methods that directly use raw spectral information and EPF input to SVM for classification, respectively. ULBP-SPG and STSE-DWLR are two spectral–spatial classification methods with superpixel segmentation for tackling small-sample classification tasks. S2DCNN, UM2L, DCFSL, DSCA-Net, GMA-Net, and DMSGer are six different deep small-sample learning methods. STSE-DWLR, S2DCNN, DSCA-Net, and DMSGer are four semisupervised methods for land-cover classification of HSIs.

For each dataset, five labeled samples in each class are randomly selected for training, and the other ones are used to learn and test. In each experiment, the performance metrics, including the overall accuracy (OA[%]), average accuracy (AA[%]), Kappa coefficient (κ), and the test and total time (seconds), are averaged by running 10 times. We also give estimates of the number of parameters for deep learning methods.

A. Dataset Description and Parameter Settings

1) Datasets: The IP HSI was captured with 200 effective spectral bands in Northwestern Indiana, including 16 classes of land covers and 145 × 145 pixels.

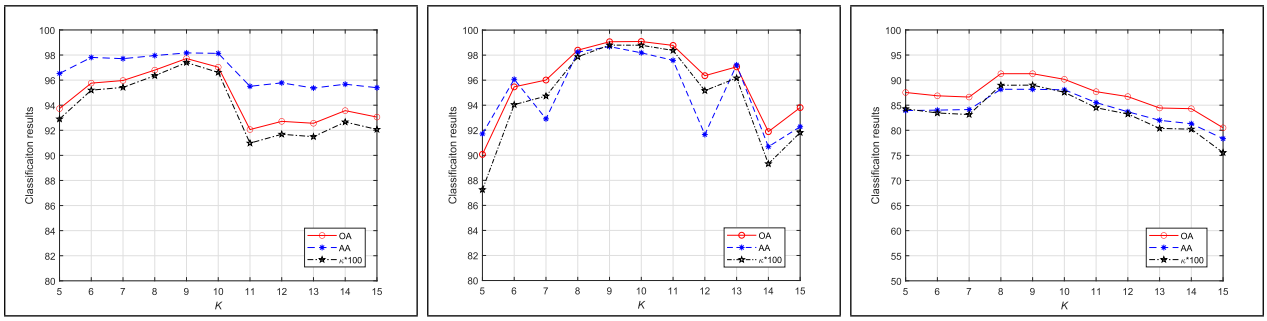
This PU HSI contains 610 × 340 pixels and 103 spectral bands, where nine classes of land covers need to be classified.

The HongHu HSI [44], [45] was collected, including 940 × 475 pixels with 270 spectral bands. This scene is a complex area with 22 classes of interest, which is a challenging dataset in HSI classification.

2) Parameter Settings: In EPFSO, the parameters w, τ, and (δ_s, δ_r) of the edge-preserving filters are set the same as these in [42]. The number of permutations K is tuned in the interval [5, 15]. In general, too few permutations K decrease the reliability of new samples, while too many permutations K reduce the number of new samples. Another important parameter is the radius of neighborhood r, which is selected from the interval [1, 6]. A larger neighborhood radius increases the time cost of the search, approximating the global search. To ensure experimental fairness, the parameters of other comparison methods refer to the corresponding references and select the most optimal results.

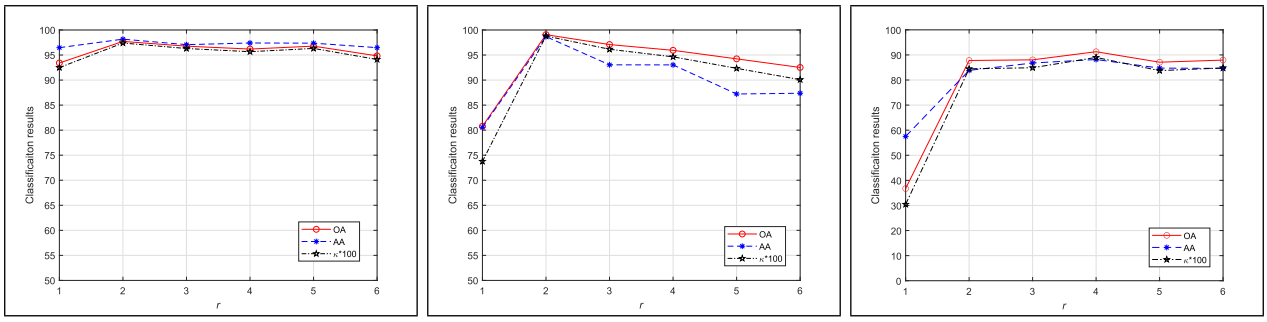
To evaluate the effect of the number of permutations K on the EPFSO method, we conducted experimental demonstration and analysis on the IP, PU, and HongHu HSIs in the case of five labeled samples in each class. Fig. 2 presents the OA, AA, and κ when choosing different parameters K on

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(a) (b) (c)

Fig. 2. OA, AA, and κ on the three HSIs for different parameters K. (a) IP. (b) PU. (c) HongHu.



(a) (b) (c)

Fig. 3. Classification accuracies for different neighborhood radii r on the three HSIs. (a) IP. (b) PU. (c) HongHu.

the three HSIs. The figures show that as K increases, the classification accuracies are also continuously improved. After reaching the peak near $K = 9$, the classification accuracies decrease, as the number of permutations K continues to grow. This can be explained by the fact that under a small number of permutations K, there are more samples learned in the AdaBoost framework, but the reliability of new samples labeled is not very high. As the number of permutations K grows, this situation will be improved. Although the number of newly labeled samples learned in the AdaBoost framework reduces, their reliability will be improved. However, as K continues to grow, the reliability continues to increase, but the number of newly labeled samples learned will gradually decrease, thus affecting the final classification performance. Thus, in the subsequent experiments, we use the number of permutations $K = 9$ for these datasets.

Furthermore, to verify the impact of the radius of neighborhood r on the EPFSO method, a similar experimental analysis is also implemented on the IP, PU, and HongHu HSIs in the case of five labeled samples in each class. Fig. 3 displays the OA, AA, and κ under different neighborhood radii r on the three HSIs. The classification results on the IP and PU HSIs reach the best when the neighborhood radius r is set to 2, i.e., the neighborhood size is 5×5, while on the HongHu HSI, they reach the best when the neighborhood radius r is set to 4, i.e., the neighborhood size is 9×9. The continuously increasing neighborhood radius leads to a gradual decrease in classification performance. Thus, the neighborhood radius $r = 2$ is chosen for the IP and PU HSIs, and $r = 4$ is chosen for the HongHu HSI.

B. Experimental Comparison

To demonstrate the superiority of EPFSO, we compared it with several HSI classification methods, including benchmark methods (SVM and PCA-EPFs), classical methods (ULBP-SPG and STSE-DWLR), and deep learning methods (S2DCNN, UM2L, DCFSL, DSCA-Net, GMA-Net, and DMSGer). Table I shows the classification performance for different methods on the IP HSI. The proposed EPFSO boasts significant advantages, whose accuracies are much higher than those of other methods in small-sample classification. Simultaneously, EPFSO achieves an acceptable time cost in terms of total computation time, significantly faster than six deep learning methods. Its test time ranked fourth out of the 11 compared methods, demonstrating its efficiency. Compared with PCA-EPFs, although both EPFSO and PCA-EPFs utilize EPF features, EPFSO achieves better classification accuracies. This illustrates the effectiveness and advancement of the proposed framework based on the self-training strategy and two screening mechanisms in newly labeled sample learning. STSE-DWLR achieves the least time, but its accuracies are unsatisfactory. Two recently proposed deep learning methods, DSCA-Net and GMA-Net, achieve better classification accuracies than eight other comparison methods, including two semisupervised deep learning methods, S2DCNN and DMSGer. Two other few-shot deep learning methods, UM2L and DCFSL, are even inferior to PCA-EPFs and STSE-DWLR in terms of classification accuracies and time cost. This further illustrates the deficiencies of deep learning methods in new sample learning, especially the time spent on model training. The method ULBP-SPG only outperforms the benchmark

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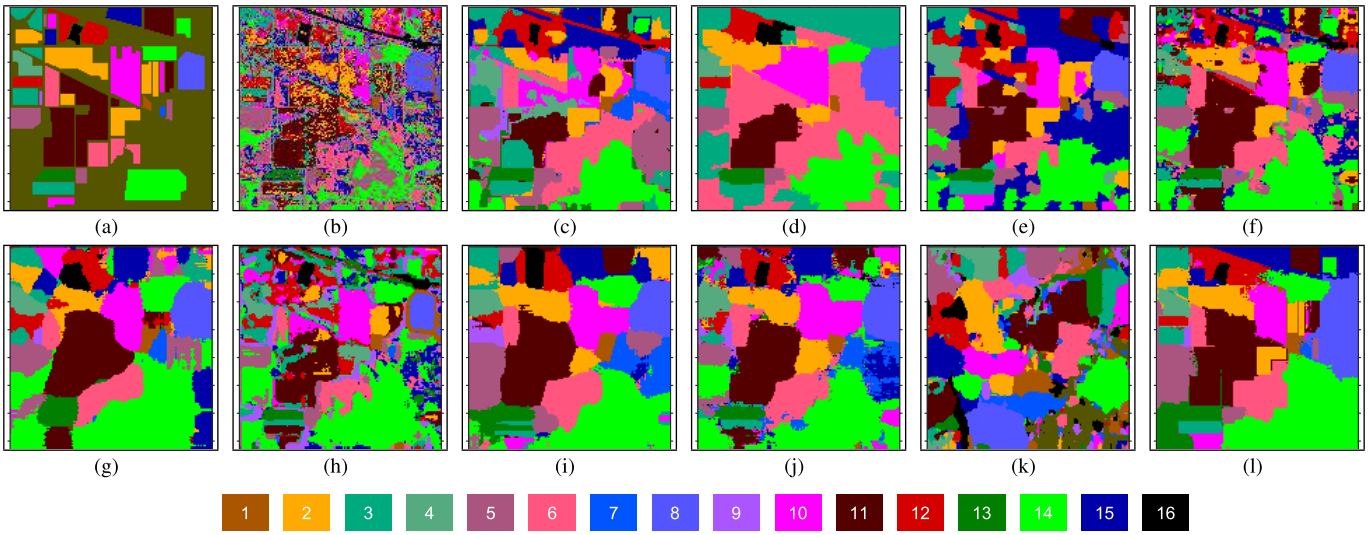


Fig. 4. Whole classification maps for different methods on the IP HSI in the case of five labeled samples in each class. (a) Ground truth. (b) SVM. (c) PCA-EPFs. (d) ULBP-SPG. (e) STSE-DWLR. (f) S2DCNN. (g) UM2L. (h) DCFSL. (i) DSCA-Net. (j) GMA-Net. (k) DMSGer. (l) EPFSO.

TABLE I

CLASSIFICATION PERFORMANCE FOR DIFFERENT METHODS WITH FIVE LABELED SAMPLES IN EACH CLASS ON THE IP HSI

Class	Samples	SVM	PCA-EPFs	ULBP-SPG	STSE-DWLR	S2DCNN	UM2L	DCFSL	DSCA-Net	GMA-Net	DMSGer	EPFSO
1	41	77.32	60.02	97.80	97.80	70.98	97.83	95.37	**100.00**	**100.00**	**100.00**	98.26
2	1423	30.41	71.55	41.81	65.33	28.38	49.16	43.26	65.66	61.55	55.72	**93.83**
3	825	40.86	48.72	56.91	68.52	14.75	59.40	57.95	46.95	35.66	52.84	**95.67**
4	232	55.09	40.00	96.81	90.60	30.69	**100.00**	80.60	94.27	91.14	**100.00**	**100.00**
5	478	44.77	80.12	73.77	89.77	27.07	58.80	72.91	89.64	62.11	68.89	**99.44**
6	725	70.65	89.21	83.71	94.12	67.19	74.66	87.96	98.33	98.08	89.83	**100.00**
7	23	89.13	34.51	70.00	95.65	83.04	**100.00**	99.57	**100.00**	**100.00**	**100.00**	97.14
8	473	80.40	**100.00**	96.64	99.05	82.11	**100.00**	86.26	99.79	**100.00**	99.87	99.79
9	15	83.33	49.97	0.00	93.33	96.67	**100.00**	99.33	**100.00**	**100.00**	**100.00**	97.50
10	967	45.26	70.78	36.20	79.28	48.53	69.44	62.44	86.69	78.19	70.27	**90.42**
11	2450	35.22	85.55	52.96	66.44	46.68	75.44	62.75	92.47	84.93	64.15	**99.89**
12	588	26.33	56.35	48.96	78.76	27.47	35.25	48.72	70.84	79.76	74.35	**99.17**
13	200	89.75	92.21	99.85	**100.00**	89.10	94.15	99.35	98.97	99.51	99.95	**100.00**
14	1260	74.20	97.97	83.84	84.56	86.06	76.13	85.40	90.76	99.21	71.57	**99.84**
15	381	29.34	74.49	76.33	99.27	24.17	95.60	66.69	**100.00**	99.48	76.30	99.74
16	88	86.93	97.67	99.09	91.82	87.73	98.92	97.61	**100.00**	**100.00**	99.77	**100.00**
OA		47.47±2.88	70.92±2.62	62.26±9.96	78.11±4.58	47.90±5.14	70.10±3.92	66.81±2.37	84.17±1.85	80.33±0.58	70.42±1.54	**97.72±0.04**
AA		59.94±2.19	71.82±3.59	69.67±7.72	87.14±2.89	56.91±2.85	80.30±2.10	77.89±0.86	84.38±3.02	81.74±0.73	82.71±0.71	**98.17±0.22**
$\kappa \times 100$		41.47±3.01	67.58±2.83	57.75±11.09	75.42±4.99	41.26±5.27	66.41±4.33	62.64±0.86	82.06±2.08	77.64±0.64	67.24±1.60	**97.39±0.05**
Test Time (s)		1.84	**0.34**	13.92	5.98	26.07	294.13	30.22	134.77	67.29	5.17	4.43
Total Time (s)		83.70	13.21	1260.90	**6.24**	3471.82	23171.50	2485.28	9566.78	8023.50	1797.83	254.56
#params		-	-	-	-	58475897	22953985	4270121	1878370	1678227	335523	-

method SVM in classification accuracies but requires a relatively high time cost.

To compare the classification effect for different methods more intuitively, we give the whole classification maps produced by all methods on the IP HSI. Referring to this ground-truth map [Fig. 4(a)], EPFSO generates the best classification map [Fig. 4(l)] with relatively clear boundaries than other comparable methods. In Fig. 4(l), except for a few samples on the boundary of two different classes that are misclassified into each other class, most of the other samples are correctly classified, which illustrates the effectiveness and advantages of EPFSO. At the same time, it can be easily analyzed that these misclassified samples are caused by the utilization of spatial information. Although some other comparable methods PCA-EPFs, ULBP-SPG, STSE-DWLR, UM2L, and DSCA-Net also produce very smooth classification maps [Fig. 4(c)–(e), (g), (i), and (k)], these classification maps suffer from varying degrees of large patches of misclassification. In particular, the classification map [Fig. 4(d)] of ULBP-SPG has two large patches of misclassification in its top-right corner. Besides, there is much salt and pepper-like noise in two maps [Fig. 4(b), (f), (h), and (j)] produced by SVM, S2DCNN, DCFSL, and GMA-Net, which shows that these methods do not use spatial information or utilize it insufficiently.

To observe the learning status of newly labeled samples in the proposed framework based on semisupervised classification under five labeled samples in each class, we show the number and accuracy of training set samples in two-stage learning for ten random trials on the IP HSI, as shown in Table II. Due to five labeled samples each class being randomly chosen for training and learning, the initial sample size is 80. In the first learning with the error-free mechanism, over 2000 newly labeled samples are introduced into the training set with almost 100% reliability for each run. Only one sample

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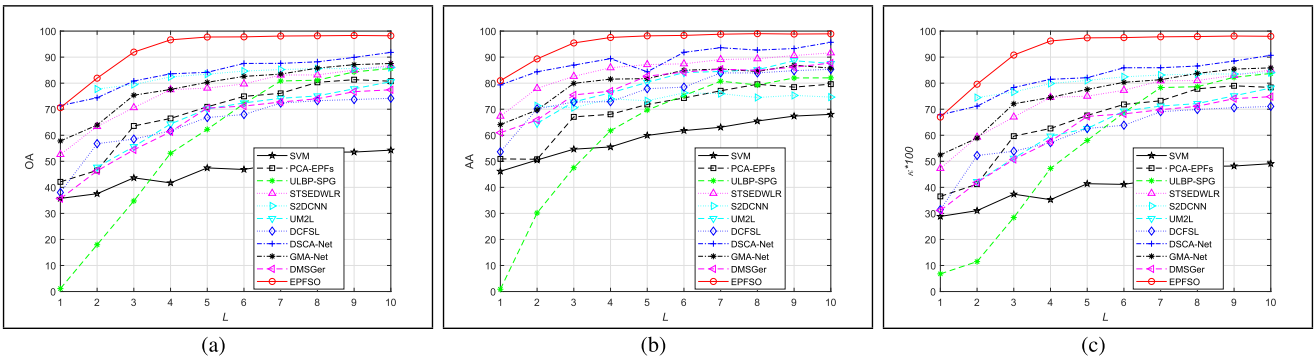


Fig. 5. Classification accuracies for different methods on the IP HSI for a different number of labeled samples L. (a) OA. (b) AA. (c) $\kappa \times 100$.

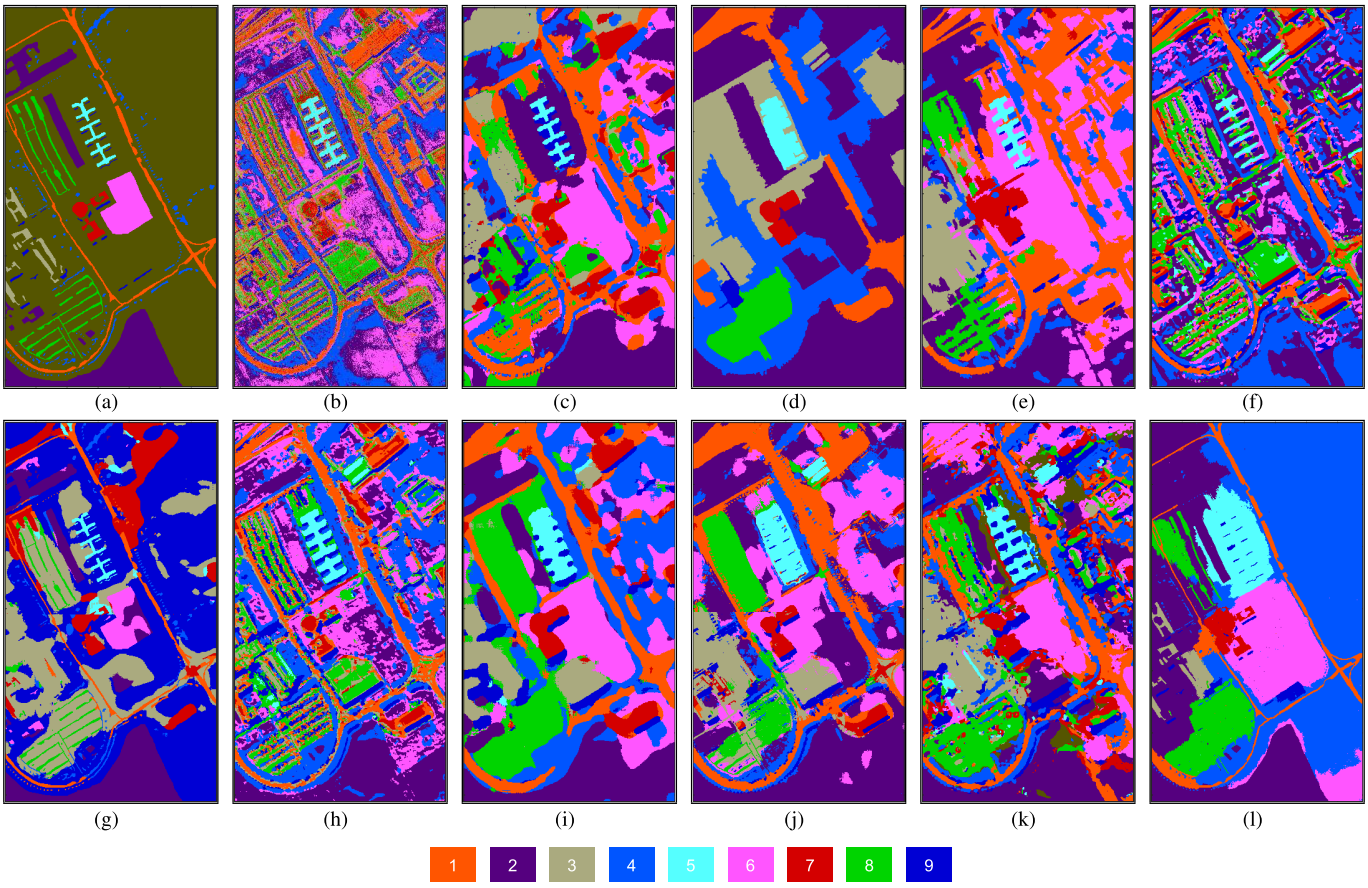


Fig. 6. Classification maps for different methods on the PU HSI in the case of five labeled samples in each class. (a) Ground truth. (b) SVM. (c) PCA-EPFs. (d) ULBP-SPG. (e) STSE-DWLR. (f) S2DCNN. (g) UM2L. (h) DCFSL. (i) DSCA-Net. (j) GMA-Net. (k) DMSGer. (l) EPFSO.

is misclassified in the 6th run. Furthermore, in the second learning by the fault-tolerant mechanism for a single error, the sample number of the training set is expanded to over 9800, but its reliability decreased to about 98.60%. By the semisupervised classification, including two-stage learning with different mechanisms, very ample training samples are learned for classifying the remaining samples. With the enlarged training set, the final classification achieves an OA of over 97.65% by the majority voting for each run.

To show the classification performance of all methods for a different number of labeled samples L, on the IP dataset, we perform ten runs by selecting 1–10 labeled sample(s) per class, respectively. Fig. 5 reports the average classification accuracies of all methods with 1–10 labeled sample(s). The results show that when only one labeled sample in each class is selected, EPFSO can achieve an OA of 70.59%, AA of 80.95%, and κ of 0.67, achieving competitive results with the DSCA-Net method. However, when the labeled sample size increases from 2 to 10, classification accuracies for all methods are improved sharply with the increase of L. The proposed EPFSO is significantly better than other methods under the same training samples in terms of OA, AA, and κ. This illustrates the advantages of EPFSO over comparison methods. When the labeled sample size L does not exceed 5, classification accuracies of EPFSO are improved sharply with an increase in L. This reflects the effectiveness of EPFSO

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TABLE II
NUMBER OF NEW SAMPLES (OA) IN DIFFERENT STAGES WITH FIVE LABELED SAMPLES IN EACH CLASS ON THE IP HSI

No.	Initial	1st learning	2nd learning	final classification
1	80	2273 (100.00)	9845 (98.59)	10249 (97.67)
2	80	2259 (100.00)	9847 (98.62)	10249 (97.72)
3	80	2240 (100.00)	9851 (98.62)	10249 (97.74)
4	80	2336 (100.00)	9841 (98.62)	10249 (97.72)
5	80	2498 (100.00)	9846 (98.58)	10249 (97.65)
6	80	2169 (99.95)	9872 (98.62)	10249 (97.74)
7	80	2394 (100.00)	9848 (98.64)	10249 (97.78)
8	80	2253 (100.00)	9847 (98.62)	10249 (97.73)
9	80	2238 (100.00)	9845 (98.64)	10249 (97.74)
10	80	2381 (100.00)	9836 (98.63)	10249 (97.75)

in small-sample learning, especially when $L < 5$. However, when the labeled sample size L exceeds 5, the classification accuracies have been improved to a very high level. Nevertheless, continuing to increase the labeled sample size L cannot always guarantee the improvement of accuracies, and there are certain fluctuations, such as an OA of 98.31%, AA of 98.87%, and κ of 0.981 in the case of nine labeled samples per class versus an OA of 98.20%, AA of 98.94%, and κ of 0.98 when ten labeled samples per class. This can be explained by the following two reasons. The first one is that enough newly labeled samples with almost 100% reliability are produced in the first learning with the error-free mechanism; refer to Table II in the case of five labeled samples in each class. Although more than five labeled samples in each class may increase the newly labeled sample size, the increase rate decreases. The second one is that the randomness of the initial labeled sample selection leads to certain fluctuations in terms of classification results when more than five labeled samples per class.

To further verify the effectiveness and advancement of EPFSO on the PU HSI, Table III displays the average classification accuracies and total time cost of different methods for ten independent runs under randomly selecting five labeled samples for each class. The proposed EPFSO achieves the best classification accuracies with an OA of 99.07%, AA of 98.68%, and κ of 0.988, which are much higher than the ones of other methods. This again demonstrates the significant advantage of the proposed EPFSO in new sample learning. Especially, compared with recently proposed deep learning methods, DSCA-Net and GMA-Net, EPFSO is significantly better than them regarding classification accuracies and time cost. In contrast, two semisupervised deep learning methods, S2DCNN and DMSGer, performed unsatisfactorily. Two other deep learning methods, DCFSL and UM2L, continue the characteristics of deep learning and require much time cost, especially UM2L. Although UM2L can achieve better classification accuracies than DCFSL, it takes more time for knowledge learning. Due to the complexity of the distribution of land covers in this HSI dataset, two spectral–spatial methods, ULBP-SPG and STSE-DWLR, do not produce very good classification results under small samples. PCA-EPFs achieve the advantage of time cost due to the low dimensionality of EPFs and outperform SVM in both classification accuracies and time cost.

To demonstrate more intuitively the advantages of EPFSO, Fig. 6 reports the whole classification maps of different methods under five labeled samples in each class. Compared with the map [Fig. 6(a)], we observe that this classification map [Fig. 6(l)] of EPFSO obtains the optimal classification result with spatial homogeneity, and only a few small dot-like areas on the left are misclassified. This is also consistent with its OA of 99.07%. This illustrates the good performance of the proposed EPFSO. Although a relatively smooth boundary is also observed in Fig. 6(c)–(e), (g), (i), and (j), some small areas are misclassified, which affects the improvement of classification accuracies. Especially, in Fig. 6(d), produced by ULBP-SPG, even some large areas are misclassified, resulting in its accuracies that are only better than SVM, but it is extremely time-consuming.

Finally, to verify the superiority of EPFSO on a newer dataset, all methods were tested on the HongHu HSI. This HSI is more complex than the other two, which is mainly reflected in the following three aspects. First, the spectral resolution is higher, and the data dimension is higher; second, the image size is larger, and the samples are larger, and third, the scene is more complex and has more classes, and many boundaries of different classes are not significantly separated, as shown in Fig. 7(a). Table IV summarizes the average classification results, test time cost, total time cost, and the number of parameters of different methods for ten independent runs under randomly selecting five labeled samples for each class. EPFSO still achieves the optimal classification accuracies. Again, it demonstrates that the proposed EPFSO boasts obvious advantages in the small-sample classification task. The recently proposed deep learning methods, DSCA-Net and GMA-Net, have achieved very high accuracies, significantly surpassing the performance of other compared methods, including two semisupervised deep learning methods, S2DCNN and DMSGer. However, due to the increase in image size, sample size, and data dimensionality in this HSI, the time cost for all methods increases significantly, especially for DCFSL, ULBP-SPG, GMA-Net, DSCA-Net, UM2L, S2DCNN, and DMSGer. Although the time cost of EPFSO also increases, it remains within an acceptable range when compared with other methods, which often suffer greatly in terms of accuracy. Although the method PCA-EPFs achieves the fastest classification, the classification accuracies are far from satisfactory. This illustrates that the features EPFs can help to quickly classify samples due to dimensionality reduction by PCA. On the other hand, it is also explained that it is difficult to achieve high-precision classification only by using the EPF features. In contrast, based on the EPF features, the proposed EPFSO produces better classification accuracies. Moreover, EPFSO can maintain significant advantages in this complex classification task.

Fig. 7 reports the classification maps for all methods by one run on this dataset. Except for SVM and DCFSL, the other methods obtain relatively smooth classification maps

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TABLE III
CLASSIFICATION PERFORMANCE FOR DIFFERENT METHODS WITH FIVE LABELED SAMPLES IN EACH CLASS ON THE PU HSI

Class	Test Samples	SVM	PCA-EPFs	ULBP-SPG	STSE-DWLR	S2DCNN	UM2L	DCFSL	DSCA-Net	GMA-Net	DMSGer	EPFSO
1	6626	44.94	76.42	52.84	77.77	82.86	73.90	82.20	93.54	90.08	72.83	**99.94**
2	18644	53.83	95.53	89.15	78.91	80.31	90.37	87.74	**99.86**	92.36	87.75	99.50
3	2094	51.26	57.22	84.62	97.30	78.53	76.18	67.46	**99.23**	94.25	79.99	95.37
4	3059	94.85	65.80	42.37	60.87	86.17	64.08	93.16	61.66	91.42	89.02	**99.00**
5	1340	96.63	89.16	89.80	97.78	89.89	73.12	99.49	**100.00**	**100.00**	98.36	**100.00**
6	5024	37.72	83.08	36.94	98.48	88.35	78.24	77.32	97.97	93.54	98.07	**99.82**
7	1325	62.38	62.45	**100.00**	97.90	84.44	96.02	81.18	99.24	99.89	90.19	98.89
8	3677	54.50	54.38	56.79	57.52	84.93	73.56	66.73	**98.18**	80.33	92.68	96.14
9	942	94.69	78.22	9.21	46.37	88.50	66.79	98.66	75.03	**99.95**	98.20	99.47
OA		55.93±7.06	76.82±3.27	69.62±12.18	79.32±6.30	78.45±4.52	81.54±7.31	83.65±1.77	95.19±1.07	91.77±1.48	87.42±4.08	**99.07±0.64**
AA		65.65±4.33	73.58±2.66	62.41±5.35	79.40±2.71	84.84±3.09	77.13±8.42	83.77±2.01	91.63±0.36	93.53±2.09	89.68±1.82	**98.68±0.66**
$\kappa \times 100$		45.87±6.60	70.56±6.60	59.33±12.63	73.48±7.52	75.20±3.18	76.12±9.04	78.70±2.01	93.61±1.39	89.26±2.13	83.79±4.41	**98.77±0.84**
Test Time (s)		4.54	**1.49**	38.43	197.63	120.80	568.36	356.44	457.33	322.63	5.39	36.43
Total Time (s)		**16.20**	24.02	662.71	224.45	31790.27	24540.26	2526.65	91318.70	68352.50	23154.90	2686.53
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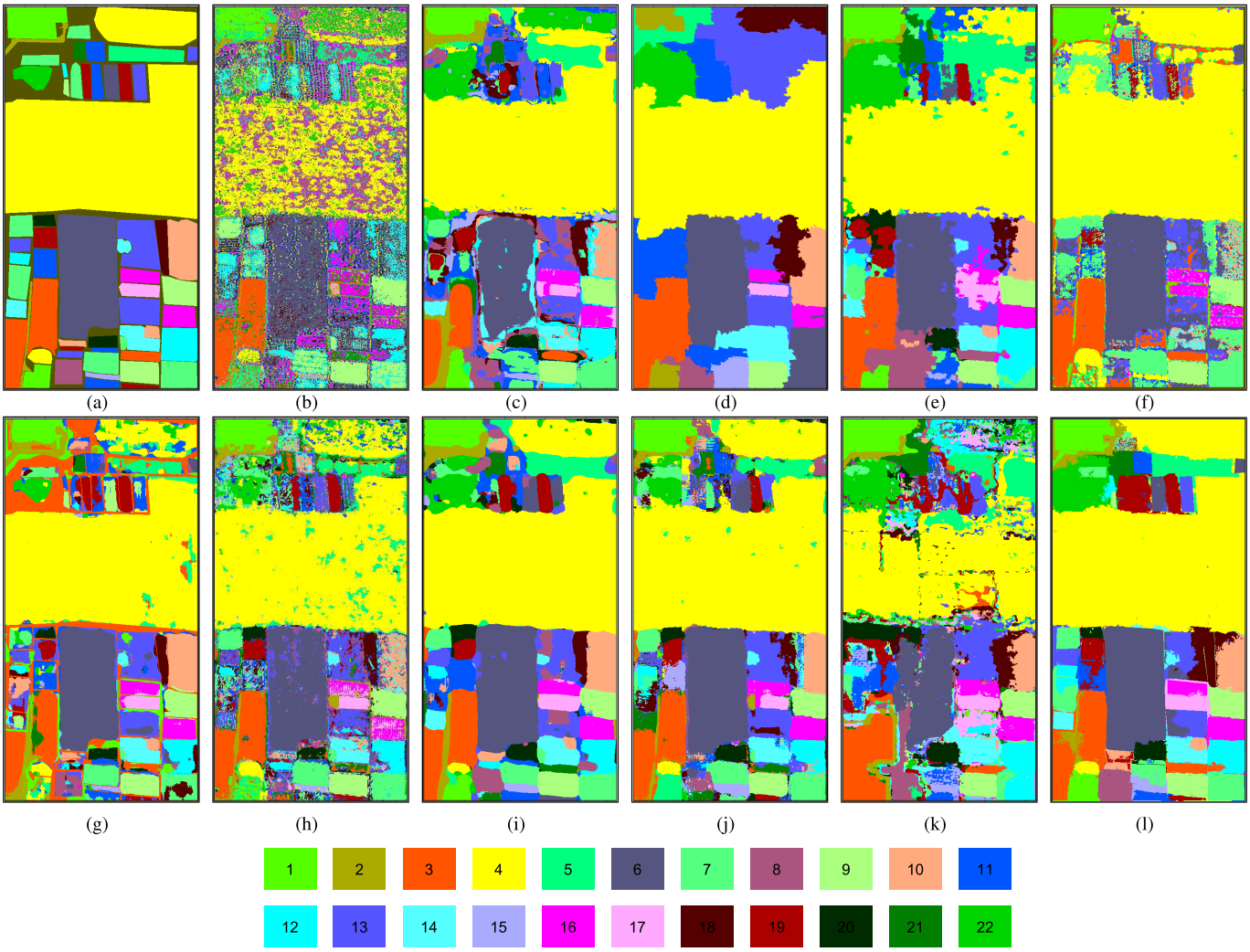


Fig. 7. Classification maps for different methods on the HongHu HSI in the case of five labeled samples in each class. (a) Ground truth. (b) SVM. (c) PCA-EPFs. (d) ULBP-SPG. (e) STSE-DWLR. (f) S2DCNN. (g) UM2L. (h) DCFSL. (i) DSCA-Net. (j) GMA-Net. (k) DMSGer. (l) EPFSO.

[Fig. 7(c)–(g) and (i)–(l)]. Although all classification maps have varying degrees of misclassification, most methods obtain good results in some large areas, especially the proposed EPFSO. Most of the large regions in Fig. 7(l) are correctly classified, except for some small-region blocks. Meanwhile, we observe that all methods do not handle boundaries very well. This is because many boundaries of different classes are not significantly separated in this dataset. This leads to the possibility of the misclassification of samples near these boundaries being very high, because it affects the effective utilization of spatial information. Nevertheless, the classification map [Fig. 7(l)] produced by EPFSO still achieves the optimal classification result visually with reference to this ground-truth map [Fig. 7(a)].

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TABLE IV
CLASSIFICATION PERFORMANCE FOR DIFFERENT METHODS WITH FIVE LABELED SAMPLES IN EACH CLASS ON THE HONGHU HSI

Class	Test Samples	SVM	PCA-EPFs	ULBP-SPG	STSE-DWLR	S2DCNN	UM2L	DCFSL	DSCA-Net	GMA-Net	DMSGer	EPFSO
1	14036	73.77	**90.94**	77.40	82.30	0.00	51.26	82.79	74.20	75.23	55.55	87.38
2	3507	70.50	40.27	17.12	47.32	72.93	94.59	**96.01**	85.59	83.97	60.34	48.19
3	21816	56.15	**93.44**	62.81	90.12	24.06	73.78	82.17	85.10	87.17	85.47	92.08
4	163280	55.12	99.34	83.94	89.01	93.18	91.68	91.12	98.13	98.69	70.02	**99.64**
5	6213	49.64	52.77	38.37	96.98	98.15	62.98	65.04	93.67	86.73	64.01	**99.88**
6	44552	49.39	96.57	71.98	88.77	9.07	87.84	81.82	93.55	97.29	76.93	**99.60**
7	24098	23.49	68.81	10.42	56.36	**93.50**	47.07	70.53	64.61	64.80	25.59	65.61
8	4049	7.65	20.56	77.62	97.60	83.38	74.37	32.13	75.39	43.81	69.84	**99.58**
9	10814	62.18	93.12	45.77	89.14	0.19	94.25	93.08	93.10	90.73	80.07	**99.70**
10	12389	10.49	75.56	47.79	77.33	85.80	**90.42**	72.69	89.14	67.00	74.95	76.10
11	11010	8.08	40.41	42.11	62.07	24.64	**67.42**	45.43	57.17	21.41	18.84	59.56
12	8949	37.16	51.99	54.12	79.38	18.43	60.79	67.42	74.18	69.79	44.46	**87.30**
13	22502	16.67	65.07	61.82	56.16	1.78	60.82	41.31	**72.28**	70.96	46.13	62.36
14	7351	7.23	68.59	29.70	78.57	48.61	72.53	65.79	90.75	68.61	54.56	**95.39**
15	997	39.50	33.79	49.95	88.32	23.42	73.85	96.09	99.30	**100.00**	94.18	98.09
16	7257	29.49	93.67	82.00	74.10	0.05	93.38	51.89	92.70	**96.36**	92.67	84.47
17	3005	45.62	59.66	89.12	99.00	65.35	88.34	71.58	**99.19**	67.74	55.01	96.98
18	3212	12.78	19.63	80.00	91.05	18.11	83.49	67.62	92.37	78.30	90.69	**99.34**
19	8707	15.96	53.25	24.10	66.58	0.00	73.14	76.41	82.76	56.89	54.60	**94.37**
20	3481	18.51	43.33	11.52	89.62	37.46	82.53	88.88	**97.86**	96.87	61.68	96.86
21	1323	24.65	29.45	10.00	99.76	0.00	80.27	50.72	97.82	80.95	**100.00**	99.25
22	4035	35.37	35.05	80.00	99.95	0.00	90.00	82.33	**99.99**	89.13	92.17	98.06
OA		44.17±5.56	77.15±1.89	66.31±11.10	82.60±3.01	74.93±0.94	81.16±4.92	79.55±2.46	89.22±1.40	86.15±0.23	65.37±3.46	**91.28±0.84**
AA		34.06±1.96	60.24±2.74	52.17±5.07	81.80±1.62	36.28±0.70	77.04±7.36	71.49±2.53	86.77±3.47	76.93±0.26	66.72±4.84	**88.17±1.20**
$\kappa \times 100$		35.62±4.95	71.86±2.10	58.47±10.39	78.62±3.42	67.71±1.22	76.64±5.82	74.73±2.87	86.47±1.76	82.50±0.27	59.04±3.34	**88.98±1.05**
Test Time (s)		72.22	**11.00**	565.37	2110.57	771.07	1281.10	619.20	1698.69	989.44	196.97	65.97
Total Time (s)		533.47	**72.21**	59809.06	850.73	63147.32	135383.32	28086.93	119972.86	93052.38	23336.59	6545.76
#params		-	-	-	-	59487172	22953985	4278087	1923310	2048615	609648	-

VI. CONCLUSION

A very efficient small-sample classification framework, EPFSO, for land-cover classification of HSIs is put forward in this work. To acquire as many high-reliability labeled samples as possible, a specialized self-training strategy is designed, along with two distinct screening mechanisms, within a novel two-stage learning process that incorporates smooth ordering, repeated learning, and interpolation classification. First, for reducing the data dimension and extracting efficient features, the edge-preserving filter is introduced for extracting low-dimensional EPF. Then, the feature embedding based on smooth ordering is built to produce a 1-D signal of EPF according to the classic shortest-path problem and neighborhood search. Next, an AdaBoost algorithm based on interpolation classification is designed in a self-training framework for learning numerous newly labeled samples with high reliability. Two distinct screening mechanisms are integrated into the repeated learning process, which aims to overcome the poor reliability of new training samples and the insufficient number of new samples. After obtaining the enlarged training set, including sufficient labeled samples, a simple majority voting method is used to classify the other samples. These high-precision classification results of land covers in three HSI scenes demonstrate the effectiveness and superiority of EPFSO in small-sample classification tasks, whose accuracies are much higher than several other comparable methods, including three recently proposed deep learning methods. Moreover, EPFSO also gains a very good performance in time cost.

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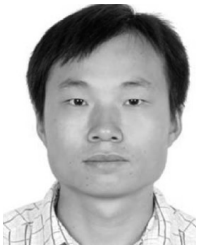
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